

No
Date

$$4.) \sum_{n=1}^{\infty} \frac{(-3)^{n+1}}{n^2} = U_n = (-3)^{n+1} n^2$$

$$\lim_{n \rightarrow \infty} \frac{3^{n+1+1}}{(n+1)^2} = \frac{3^{n+2}}{3^{n+1} n^2}$$

$$= \frac{3^{n+2}}{3^{n+1} n^2} \times \frac{n^2}{n^2} = \frac{3^{n+2}}{3^{n+1} n^2} = \frac{3}{n^2/n^2 + 2n/n^2 + 1/n^2} = \frac{3}{1 + 0 + 0} = \frac{3}{1} = 3$$

yang mutlak 3 (konvergen mutlak)

yang tidak mutlak -3 (konvergen bersyarat)

$$1.) \sum_{n=1}^{\infty} \frac{n}{2^n} = \lim_{n \rightarrow \infty} \frac{n+1}{\frac{2^{n+1}}{2^n}}$$

$$= \frac{n+1}{2^{n+1}} \times \frac{2^n}{n} = \frac{1+\frac{1}{n}}{2} = \frac{1+0}{2} = \frac{1}{2} \quad (\text{konvergen mullak})$$

$$2.) \sum_{n=1}^{\infty} \frac{n^2}{e^n} = \lim_{n \rightarrow \infty} \frac{\frac{(n+1)^2}{e^{n+1}}}{\frac{n^2}{e^n}}$$

$$= \frac{(n+1)^2}{e^{n+1}} \times \frac{e^n}{n^2} = \frac{n^2+2n+1}{e^{n+1}} \times \frac{e^n}{n^2}$$

$$= \frac{n^2+2n+1}{e^{n+1}} : n^2 = \frac{1+2n/n^2+1/n^2}{e} = \frac{1}{e} = \frac{1}{2.7} \quad (\text{konvergen mullak})$$

$$3.) \sum_{n=1}^{\infty} \frac{n^4}{2^n} = \lim_{n \rightarrow \infty} \frac{\frac{(n+1)^4}{2^{n+1}}}{\frac{n^4}{2^n}}$$

$$= \frac{(n+1)^4}{2^{n+1}} \times \frac{2^n}{n^4}$$

$$= \frac{n^4+4n^3+1}{2^{n+1}} \times \frac{2^n}{n^4} = \frac{n^4+4n^3+1}{2n^4} : n^4$$

$$= \frac{1+4n/n^4+1/n^4}{2} = \frac{1+0+0}{2} = \frac{1}{2} \quad (\text{konvergen mullak})$$